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Shenfu injection reduces toxicity of bupivacaine in rats

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Objective To investigate the effects of injecting Shenfu, an extract of traditional Chinese herbal medicines, on the central nervous system (CNS) and the cardiac toxicity of bupivacaine in rats.

Methods Sixteen male Sprague-Dawley rats, weighing form 280 to 320 g, were randomly assigned to two groups (n=8 in each group). Animals in the control group received a saline injection 10 ml/kg while animals in the Shenfu group received an injection of Shenfu 10 ml/kg intraperitoneally 30 minutes before intravenous infusion of bupivacaine. Lead Ⅱ of an electrocardiogram (EEG) was continuously monitored after 3 needles were inserted into the skin of both forelimbs and the left hind-leg of each rat. The femoral artery was cannulated for measurement of arterial blood pressure and blood sampling. The femoral vein was cannulated for the infusion of bupivacaine. After baseline measurement (arterial blood pressure, heart rate and arterial blood gas), 0.5% bupivacaine was infused intravenously at a rate of 2 mg · kg^{-1} · min^{-1} to all animals until asystole occurred. The time of bupivacaine-induced convulsions, arrhythmia and asystole were determined. The dose of bupivacaine was then calculated at the corresponding time point.

Results The doses of bupivacaine that induced convulsions, arrhythmia and cardiac arrest were remarkably larger in Shenfu injection-treated animals than in saline-treated rats [convulsions, (10.5±1.9) mg/kg vs (7.2±1.5) mg/kg; arrhythmia (10.5±2.0) mg/kg vs (7.2±1.9) mg/kg; asystole, (32.8±8.5) mg/kg vs (25.0±5.0) mg/kg; *P*=0.006, 0.009 and 0.044, respectively].

Conclusion The Shenfu injection is able to reduce the toxicity of bupiralaine to CNS and cardiac system in rats.

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Bupivacaine is widely used as a long-acting local anesthetic in clinical anesthesia. However, bupivacaine has been reported linking to unreversed central nervous system (CNS) and cardiac toxicity when a patient receives an over-dose or this drug is administered intravascularly by accident.[1] It is important in clinical anesthesia to prevent and alleviate the CNS and cardiac toxicity of bupivacaine.[2] Shenfu, an extract of traditional Chinese herbs, is effective in treating cardiac diseases including coronary heart disease, cardiac arrhythmia and congestive heart failure. Moreover, some studies have proved that Shenfu injection has significant protective effects against ischemia-reperfusion injuries of many organs such as the brain, spinal cord, heart, liver and kidney.[3,4] The present study was undertaken to investigate whether or not Shenfu injection could reduce the CNS and cardiac toxicity of bupivacaine in rats.

METHODS

Animals and group assignment

The animals were provided by the Experimental Animal Center of the Fourth Military Medical University. Sixteen male Sprague-Dawley rats (280—300 g) were randomly assigned to two groups (n=8 in each group): control and Shenfu groups. Animals in the control group received

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saline 10 ml/kg while animals in the Shenfu group received Shenfu injection 10 ml/kg intraperitoneally 30 minutes before intravenous infusion of bupivacaine. Bupivacaine was supplied by Shanghai Harvest Pharma-ceutical. Co., Ltd (No. 20020626); Shenfu injection was made by Ya'an Sanjiu Pharmaceutical. Co., Ltd (No. 020103). The main components of Shenfu injection include *Ginseng Saponin* and *Aconitine*.

Monitoring of physiological variables and infusion of bupivacaine

After overnight starvation with unrestricted access to water, the rats were placed in a plastic box with 4% isoflurane and 96% oxygen. Animals were allowed to breath freely . Anesthesia was maintained by inhalation of 2% isoflurane in oxygen administered by a face mask at a flow rate of 2 L/min. With the rats in the supine position, body temperature was continuously monitored with a flexible probe (Spacelab, USA) inserted 5 cm deep into the rectum and core temperature was maintained at 37℃ — 37.5℃ by heating lamps. Lead Ⅱ of an electrocardiogram (EEG) was continuously monitored via 3 needles that were inserted into the skin of both forelimbs and the left hind-leg. The femoral vein was cannulated with a 24-gauge catheter for the infusion of bupivacaine, and the femoral artery was cannulated with a 24-gauge catheter for continuous monitoring of arterial blood pressure and for blood sampling to determine the levels of blood gases (PaO_2, pH, SaO_2 and $PaCO_2$). After the basic physiological variables (arterial blood pressure, heart rate and body temperature) were recorded and a blood sample was drawn for blood gases, 0.5% bupivacaine was infused intravenously at a rate of 2 $mg \cdot kg^{-1} \cdot min^{-1}$ until asystol occurred. Isoflurane inhalation was stopped just before the infusion of bupivacaine. Arterial blood pressure, heart rate and ECG were continuously monitored. The time of the occurring of bupivacaine-induced convulsions, arrhythmia and asystole were determined. The doses of bupivacaine were calculated at the corresponding time point. When convulsion or arrhythmia occurred, blood samples were drawn again.

Statistical analysis

All data are expressed as mean ± SD. Statistical differences were analyzed with Student's *t* test following Levene's test. A *P* value <0.05 was considered statistically significant.

RESULTS

Blood gas analysis

The results of blood gas analysis are shown in Table 1. There were no statistical differences in the pH, PaO_2, $PaCO_2$ and SaO_2 levels between the two groups before the infusion of bupivacaine. After infusion of bupivacaine, the PaO_2 and SaO_2 levels of the two groups were similar. However, the pH of the Shenfu group was markedly lower than that of the control group ($P<0.01$), and the $PaCO_2$ level of the Shenfu group was significantly higher than that of the control group ($P<0.01$) when convulsion occurred.

Hemodynamics

The mean arterial blood pressure and heart rate were similar in the two groups (Figs. 1 and 2).

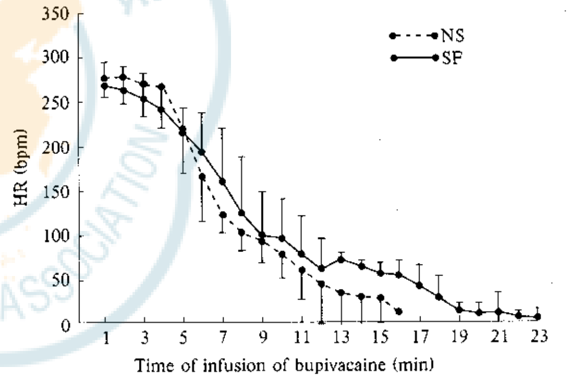

Fig. 1. The mean arterial blood pressure (MAP) in the two groups during infusion of bupivacaine (mean±SD, n=8 in each group).

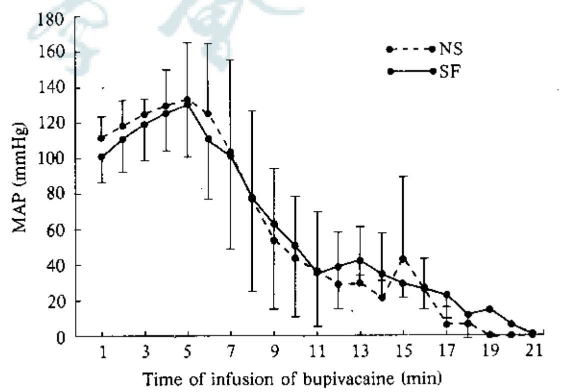

Fig. 2. The HR in the two groups during infusion of bupivacaine (mean±SD, n=8 in each group).

Table 1. Blood gas in two groups (mean±SD, n=8 in each group)

Groups	pH		PaO_2 (mmHg)		$PaCO_2$ (mmHg)		SaO_2 (%)	
	Basline	Convulsion	Basline	Convulsion	Basline	Convulsion	Basline	Convulsion
Control	7.35±0.03	7.27±0.12	44.27±7.01	35.35±10.54	5.36±0.82	7.30±2.27	99.79±0.11	99.24±0.79
Shenfu	7.38±0.04	7.25±0.09*	45.2±6.17	43.78±7.40	5.08±0.66	7.74±1.84**	99.8±0.00	99.73±0.11

* $P<0.05$, ** $P<0.01$ vs control group.

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Doses of bupivacaine induced toxicity

The cumulative doses of bupivacaine that induced convulsion, arrhythmia and asystole in the Shenfu group were remarkably larger than those in the control group (Table 2).

Table 2. Doses of bupivacaine induced toxicity in the two groups (mean±SD, n=8 in each group)

Toxic symptom	Control (mg/kg)	Shenfu (mg/kg)	*P* value
Convulsion	7.2±1.5	10.5±1.9**	0.006
Arrhythmia	7.2±1.9	10.5±2.0**	0.009
Asystole	25.0±5.0	32.8±8.5*	0.044

* $P<0.05$, ** $P<0.01$ vs control group.

DISCUSSION

The present study demonstrated that 10 ml/kg of Shenfu injected intraperitoneally before intravenous infusion of bupivacaine could postpone bupivacaine-induced convulsion, arrhythmia and asystole. It also suggested that Shenfu injection could reduce the CNS and cardiac toxicity of bupivacaine in rats.

In the present study, the mean arterial blood pressure and heart rate of the rats in the two groups during infusion of bupivacaine was consistent. When convulsion occurred, the pH of the Shenfu group was lower than that of the control group ($P<0.05$) while the level of $PaCO_2$ in the Shenfu group was higher than that of the control group ($P<0.01$). The possible explanation of this result is that the time of persistence from infusion of bipuvacaine to convulsion in the Shenfu group was longer than that in the control group, so there was a more serious compensated respiratory acidosis existing in the Shenfu group.

Bupivacaine is still widely used as a long-acting local anesthetic in clinical anesthesia. CNS and cardiac toxicity of bupivacaine resulting in convulsion and cardiovascular collapse accompanying severe arrhythmia or cardiac arrest are rare but fatal.[1] Therefore, it is very important to study any drugs that could prevent and alleviate bupivacaine toxicity in clinical anesthesia.[2] The possible mechanisms resposnsible for bupivacaine toxicity include: ① Bupivacaine's fast block of sodium and potassium channels within milliseconds of application but slow recovery from the block could result in the increase of extracellular sodium and potassium leading to functional conduction block. Moreover, bupivacaine significantly prolongs the ventricular effective refractory period and slows longitudinal and transverse conduction velocity in a dose-dependent manner. ② Bupivacaine might induce calcium release but inhibit calcium uptake, which results in the increase of extracellular calcium.[8] A number of studies have shown that two calcium channel-blocking drugs (verapamil and nimodipine) are effective in decreasing bupivacaine cardiotoxicity.[9] ③ In anesthetized patients, propofol and sevoflurane attenuate bupivacaine-induced CNS and cardiovascular toxicity.[10] The effects may be associated to the inhibition of the CNS produced by the general anesthetics. ④ β-estradiol is able to increase the bupivacaine toxicity[11] while nitric oxide (NO) decreases bupivacaine toxicity.[12,13]

The main components of Shenfu injection include Ginseng saponin and Aconitine. Shenfu injection is effective in treating chronic cardiac dysfunction and countershock, and is able to improve cases of cardiac arrhythmia, e.g. premature supraventricular or ventricular beats. Shenfu injection is reportedly able to strengthen myocardium contractility, modulate cardiac rhythm, decrease peripheral vascular resistance and to improve the microcirculation by mediating CNS function. Our previous study has demonstrated that Shenfu injection significantly protects against cerebral and spinal cord ischemia-reperfusion injuries.[4,5] Therefore, we designed the present study to verify whether or not Shenfu injection could significantly induce the tolerance to CNS and cardiac toxicity induced by bupivacaine in rats. Our results have demonstrated that Shenfu injection could postpone convulsion, arrhythmia and asystole of rats, and attenuate the bupivacaine toxicity.

The present study showed that the mean dose of bupivacaine that induced the arrhythmia of rats was 7.2 mg/kg, which is different from previous reports of 5.1 mg/kg[12] and 12.4 mg/kg.[14] This discrepancy might be due to the various anesthetics used in different studies. In our experiment, all animals were anesthetized according to an identical anesthetic procedure, and the results were still comparable.

It remains unclear why pretreatment with Shenfu injection could reduce the bupivacaine-induced CNS and cardiac toxicity. According to the studies related to organ protection of Shenfu injection, Shenfu injection can act via different pathways, levels and targets simultaneously. The results of the present study have some clinical significance. Further research is needed to ascertain whether or not Shenfu injection could be used more effectively in patients who are more sensitive to bupivacaine toxicity.[15] However, more information about Shenfu induced tolerance to toxicity of bupivacaine must be collected before this can be advocated.

In conclusion, the present study showed that Shenfu injection is able to reduce bupivaicaine-induced CNS and cardiac toxicity in rats.

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